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EE115 Analog Circuits 模拟电路

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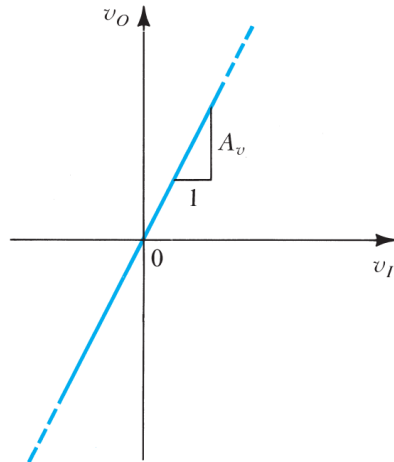
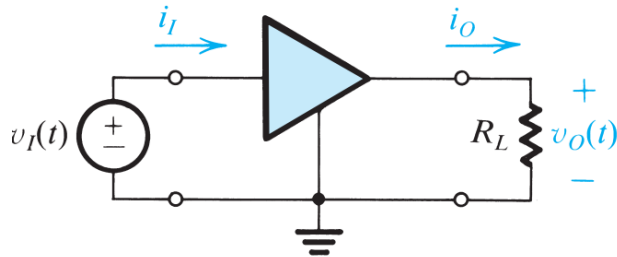
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Outline



- OP AMP Basics
- Sedra/Smith Book Page 15-43

Amplifier Gain (增益)



- Voltage Gain:

$$A_v = \frac{v_O}{v_I}$$

- Current Gain:

$$A_i = \frac{i_O}{i_I}$$

- Power Gain:

$$A_p = \frac{p_L}{p_I} = \frac{v_O i_O}{v_I i_I}$$
$$A_p = A_v A_i$$

- Note: A_v and A_i can be **positive**, **negative**, or even **complex** numbers. Negative gain means the output is 180° out of phase with input. However, A_p should always be a **positive** number.

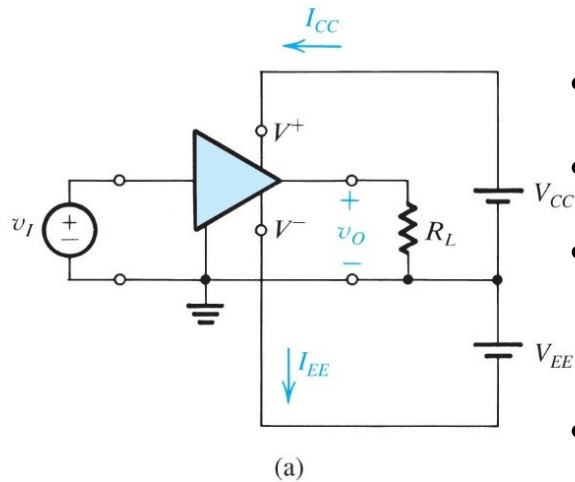
- Gain is usually expressed in Decibel (dB, 分贝)

$$A_v(dB) = 10 \log |A_v|^2 = 20 \log |A_v|$$

$$A_i(dB) = 10 \log |A_i|^2 = 20 \log |A_i|$$

$$A_p(dB) = 10 \log |A_p|$$

Amplifier Power Supply and Dissipation



- Circuit needs dc power supplies to function
- Typical power supplies are denoted as V_{CC} and $-V_{EE}$
- Total dc power dissipation $P_{dc} = V_{CC}I_{CC} + V_{EE}I_{EE}$

- Power balance $P_{dc} + P_I = P_L + P_{diss}$

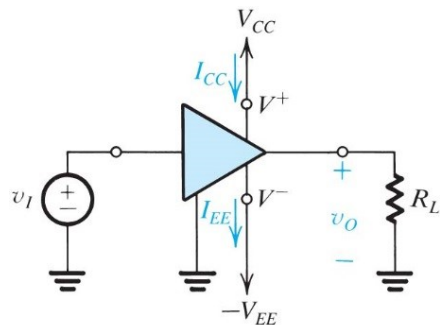


Figure 1.13 An amplifier that requires two dc supplies (shown as batteries) for operation.

- P_i : power drawn from signal source
- P_L : power delivered to the load
- P_{diss} : power dissipated in the amplifier circuit
- Power efficiency $\eta = \frac{P_L}{P_{dc}}$

– Important for power amplifiers: stereo output, transmitters

Amplifier Power Supply and Dissipation



Example 1.2

Consider an amplifier operating from $\pm 10\text{-V}$ power supplies. It is fed with a sinusoidal voltage having 1 V peak and delivers a sinusoidal voltage output of 9 V peak to a $1\text{-k}\Omega$ load. The amplifier draws a current of 9.5 mA from each of its two power supplies. The input current of the amplifier is found to be sinusoidal with 0.1 mA peak. Find the voltage gain, the current gain, the power gain, the power drawn from the dc supplies, the power dissipated in the amplifier, and the amplifier efficiency.

Solution

$$A_v = \frac{9}{1} = 9 \text{ V/V}$$

or

$$A_v = 20 \log 9 = 19.1 \text{ dB}$$

$$\hat{I}_o = \frac{9 \text{ V}}{1 \text{ k}\Omega} = 9 \text{ mA}$$

$$A_i = \frac{\hat{I}_o}{\hat{I}_i} = \frac{9}{0.1} = 90 \text{ A/A}$$

or

$$A_i = 20 \log 90 = 39.1 \text{ dB}$$

$$P_L = V_{o\text{rms}} I_{o\text{rms}} = \frac{9}{\sqrt{2}} \frac{9}{\sqrt{2}} = 40.5 \text{ mW}$$

$$P_I = V_{i\text{rms}} I_{i\text{rms}} = \frac{1}{\sqrt{2}} \frac{0.1}{\sqrt{2}} = 0.05 \text{ mW}$$

$$A_p = \frac{P_L}{P_I} = \frac{40.5}{0.05} = 810 \text{ W/W}$$

or

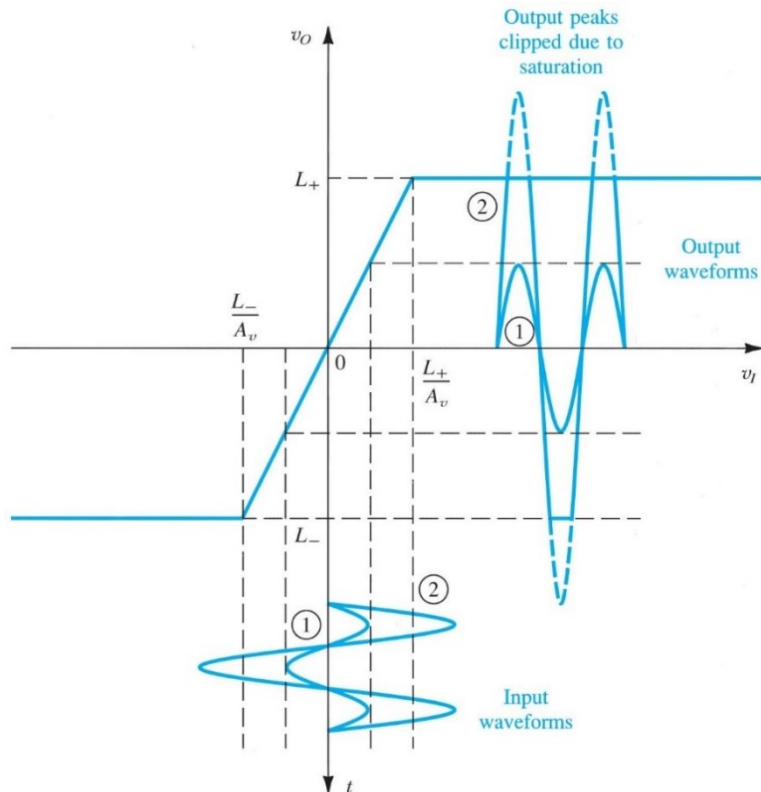
$$A_p = 10 \log 810 = 29.1 \text{ dB}$$

$$P_{\text{dc}} = 10 \times 9.5 + 10 \times 9.5 = 190 \text{ mW}$$

$$\begin{aligned} P_{\text{dissipated}} &= P_{\text{dc}} + P_I - P_L \\ &= 190 + 0.05 - 40.5 = 149.6 \text{ mW} \end{aligned}$$

$$\eta = \frac{P_L}{P_{\text{dc}}} \times 100 = 21.3\%$$

Linear Range vs. Saturation(饱和) Range



- Within linear range, the output voltage (or current) is **proportional** to the input voltage (or current)
- Beyond **linear range**, the output voltage (or current) waveforms **saturates**, resulting in distortions
 - Lose fidelity (失真) in stereo system

Figure 1.14 An amplifier transfer characteristic that is linear except for output saturation.

Symbol Notation



$$i_c(t) = I_C + i_c(t)$$

- where, $i_c(t)$
 - Lowercase-uppercase-> **total** current
- I_C
 - Uppercase-uppercase-> **dc** current
- $i_c(t)$
 - Lowercase-lowercase -> **small signal ac** current

$$i_c = I_c \sin(\omega t)$$

- where, I_c
 - Uppercase-lowercase-> **amplitude of ac** current (sinwave)

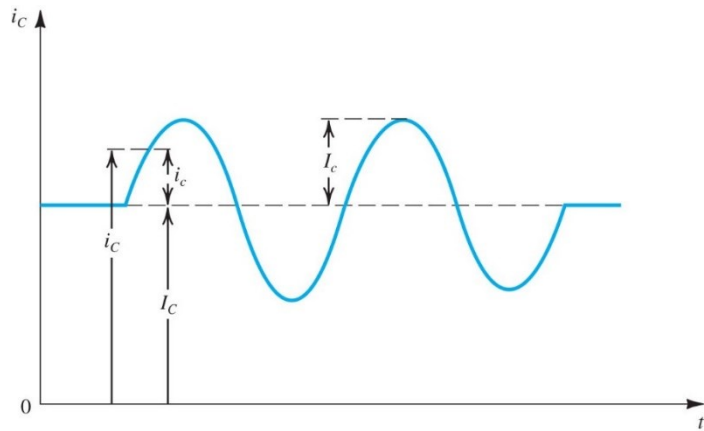


Figure 1.15 Symbol convention employed throughout the book.

Circuit Model of Voltage Amplifiers

(Two-Port Model)

- A_{vo} : open-circuit voltage gain
- R_i : input resistance of the amplifier
- R_o : output resistance of the amplifier
- R_s : source resistance
- R_L : load resistance

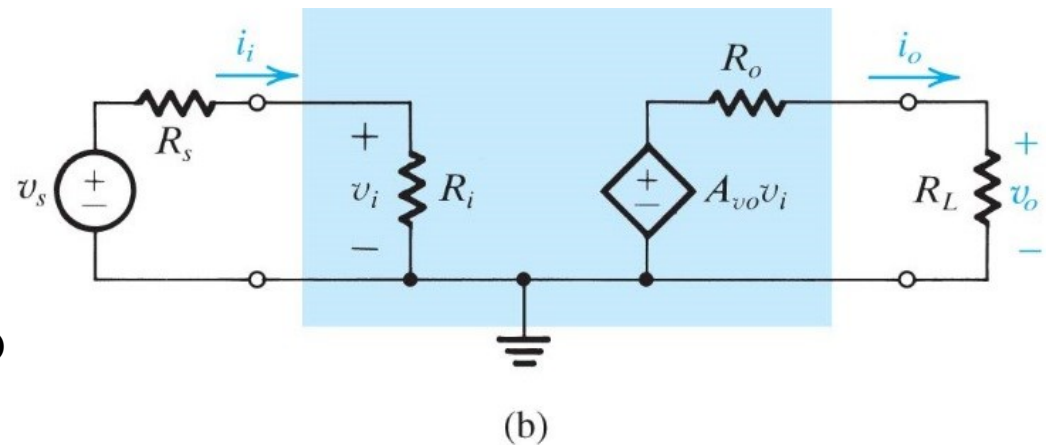
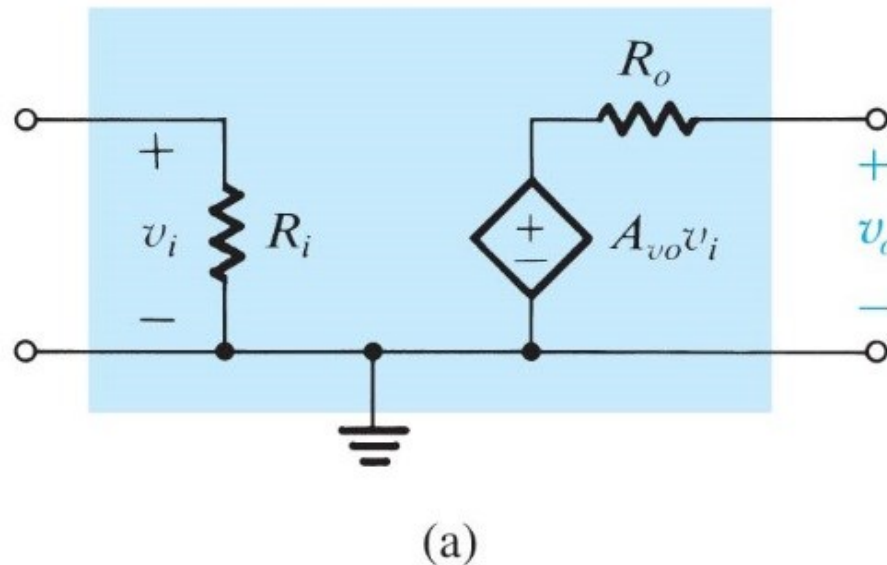


Figure 1.16 (a) Circuit model for the voltage amplifier. (b) The voltage amplifier with input signal source and load.

Cascaded Amplifier

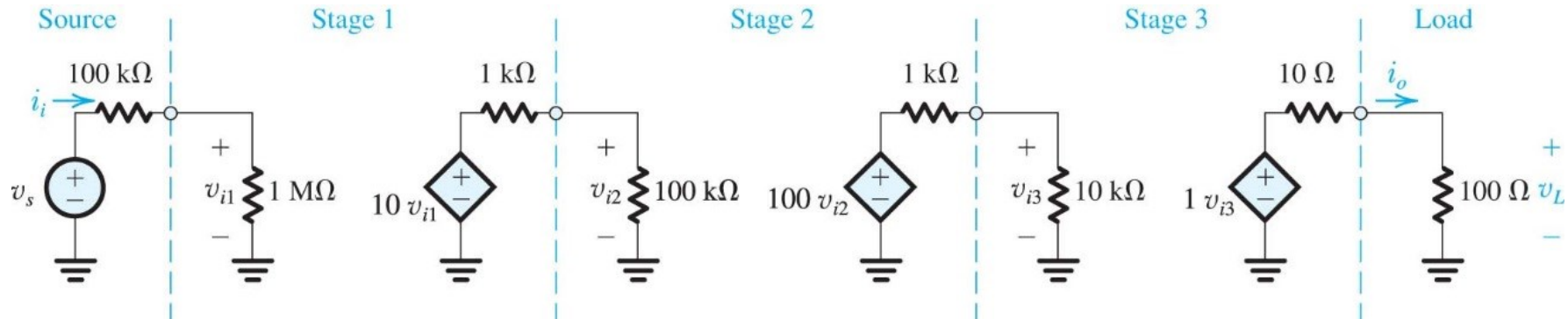


Figure 1.17 Three-stage amplifier for Example 1.3.

For most practical applications, multiple stages of amplifiers are cascaded to

1. Provide sufficient **gain**
2. Provide adequate input and output **resistances/impedances**

For example, in a voltage amplifier

- The **input stage** is designed to have **high input impedance**
- The **output stage** is designed to have **low output impedance**
- **Middle stage** provides the necessary **gain**

Amplifier Types



Table 1.1 The Four Amplifier Types

Type	Circuit Model	Gain Parameter	Ideal Characteristics
Voltage Amplifier		Open-Circuit Voltage Gain $A_{vo} \equiv \frac{v_o}{v_i} \Big _{i_o=0} \text{ (V/V)}$	$R_i = \infty$ $R_o = 0$
Current Amplifier		Short-Circuit Current Gain $A_{is} \equiv \frac{i_o}{i_i} \Big _{v_o=0} \text{ (A/A)}$	$R_i = 0$ $R_o = \infty$
Transconductance Amplifier		Short-Circuit Transconductance $G_m \equiv \frac{i_o}{v_i} \Big _{v_o=0} \text{ (A/V)}$	$R_i = \infty$ $R_o = \infty$
Transresistance Amplifier		Open-Circuit Transresistance $R_m \equiv \frac{v_o}{i_i} \Big _{i_o=0} \text{ (V/A)}$	$R_i = 0$ $R_o = 0$

Depending on the nature of the source signals and output loads, different types of amplifiers are needed:

- Voltage amplifier
- Current amplifier
- Transconductance (跨导) amplifier
 - * voltage to current
- Transimpedance (跨阻) amplifier
 - * current to voltage

Amplifier Frequency Response

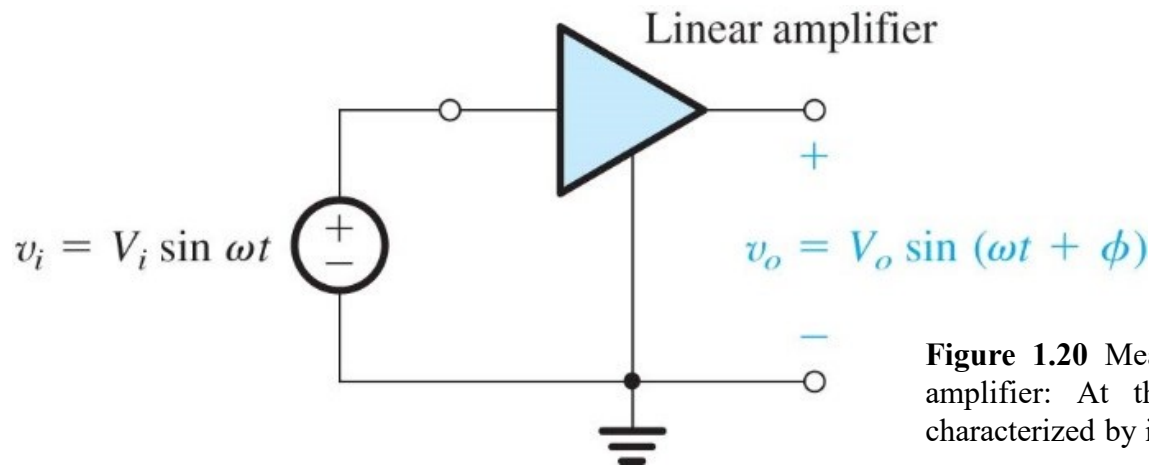


Figure 1.20 Measuring the frequency response of a linear amplifier: At the test frequency, the amplifier gain is characterized by its magnitude (V_o/V_i) and phase ϕ .

When a sinusoidal signal is applied to a linear amplifier, the output is sinusoidal:

- With the same frequency as the input
- But different **amplitude (幅度)** and **phase (相位)**

Transfer function (传递函数):

$T(\omega) = V_o(\omega) / (V_i(\omega))$

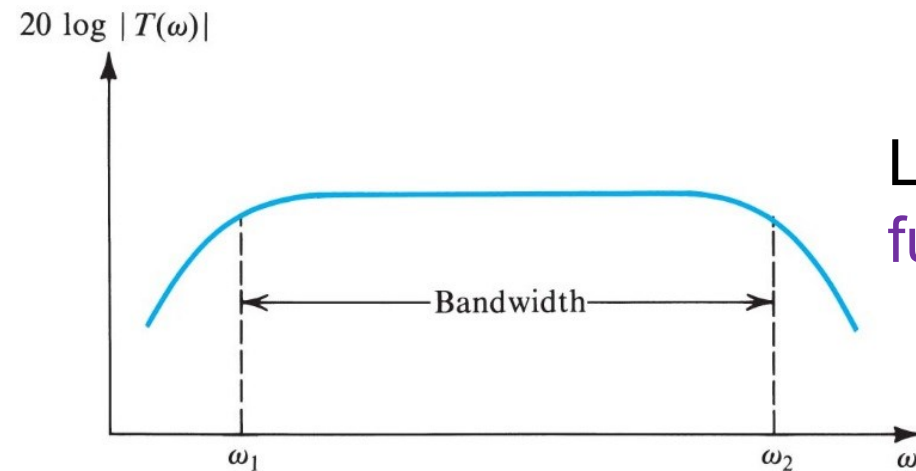
Amplitude response

$|T(\omega)| = V_o / V_i$

Phase response

$\angle T(\omega) = \phi$

Frequency Response



Log-log plot of the **transfer function** vs. **angular frequency**, ω

- Vertical axis: $20\log |T(\omega)|$

- Horizontal axis: $10\log \omega$

This is called **Bode Plot(波特图)**

Bandwidth(带宽)

- Band of frequencies over which the gain response falls by 3dB

Figure 1.21 Typical magnitude response of an amplifier: $|T(\omega)|$ is the magnitude of the amplifier transfer function—that is, the ratio of the output $V_o(\omega)$ to the input $V_i(\omega)$.

Frequency Response of Low-Pass Filters

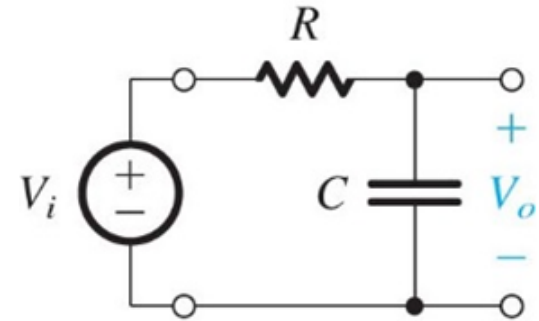
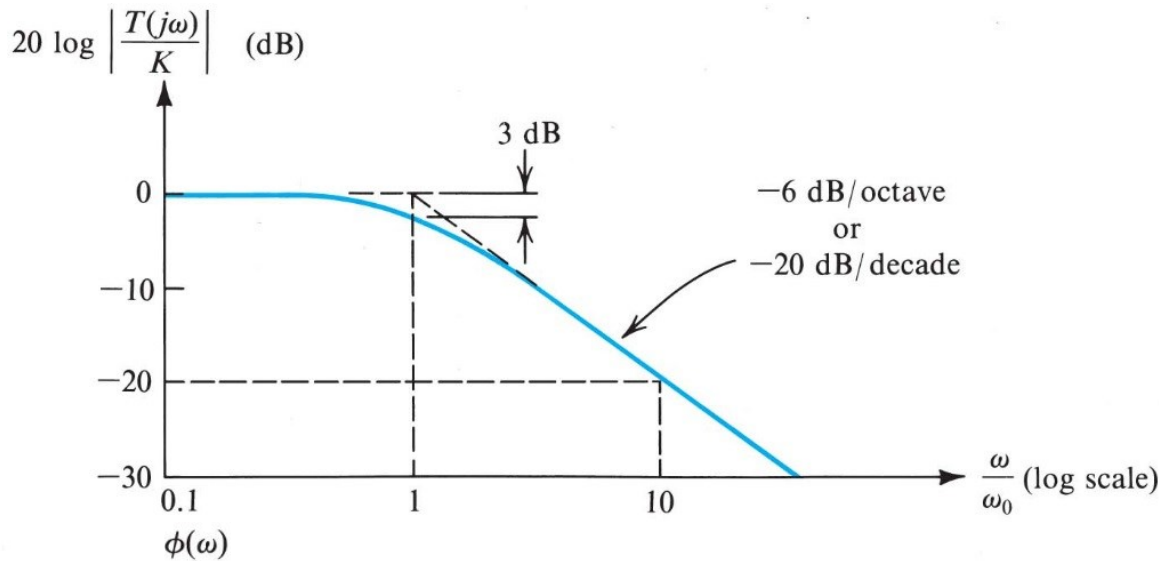


Figure 1.22 (a) a low-pass network.

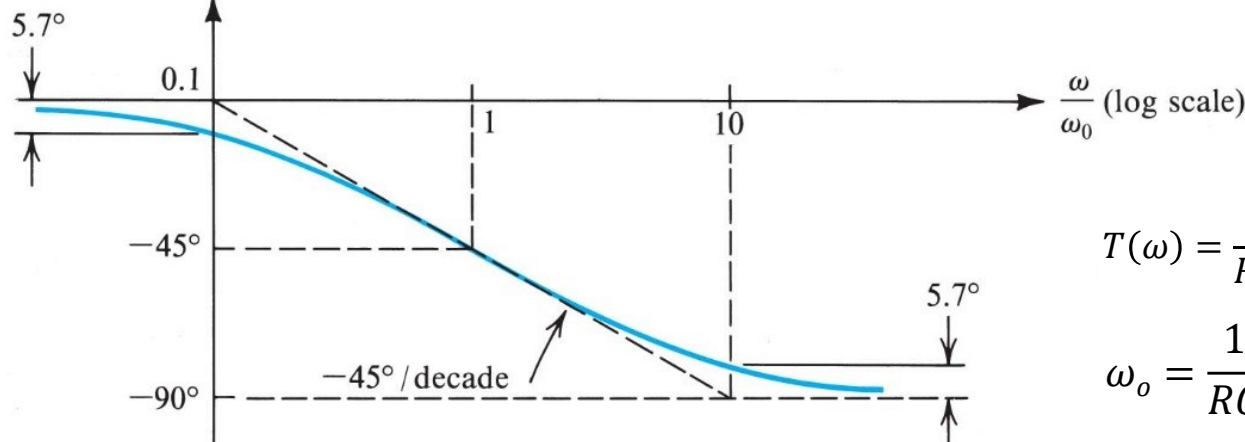


Figure 1.23 (a) Magnitude and (b) phase response of STC networks of the low-pass type.

Transfer function:

$$T(\omega) = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC} = \frac{1}{1 + j\omega/\omega_0}$$

$$\omega_0 = \frac{1}{RC}$$

$$|T(\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_0)^2}}$$

$$\angle T(\omega) = -\tan^{-1}(\omega/\omega_0)$$

$$\omega_{3dB} = \omega_0$$

Frequency Response of High-Pass Filters

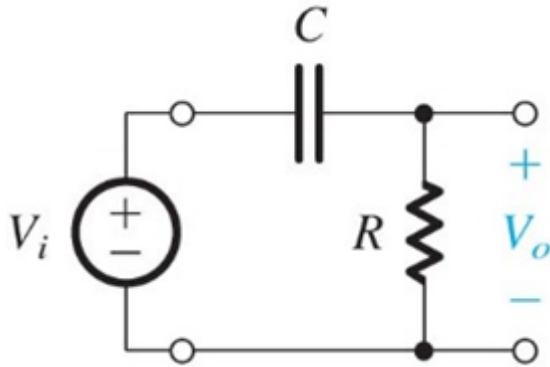
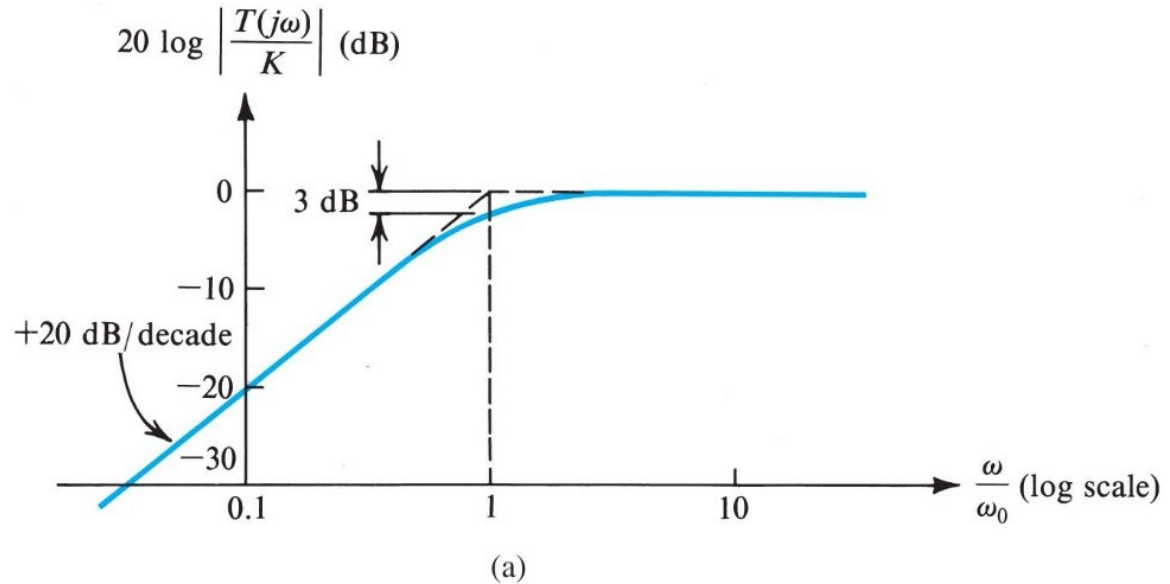


Figure 1.22 (a) a high-pass network.



Transfer function:

$$T(\omega) = \frac{R}{R + 1/(j\omega C)} = \frac{1}{1 + 1/(j\omega RC)}$$

$$= \frac{1}{1 - j\omega_o/\omega}$$

Where, $\omega_o = \frac{1}{RC}$

$$|T(\omega)| = \frac{1}{\sqrt{1 + (\omega_o/\omega)^2}}$$

$$\angle T(\omega) = \tan^{-1}(\omega_o/\omega)$$

$$\omega_{3dB} = \omega_o$$

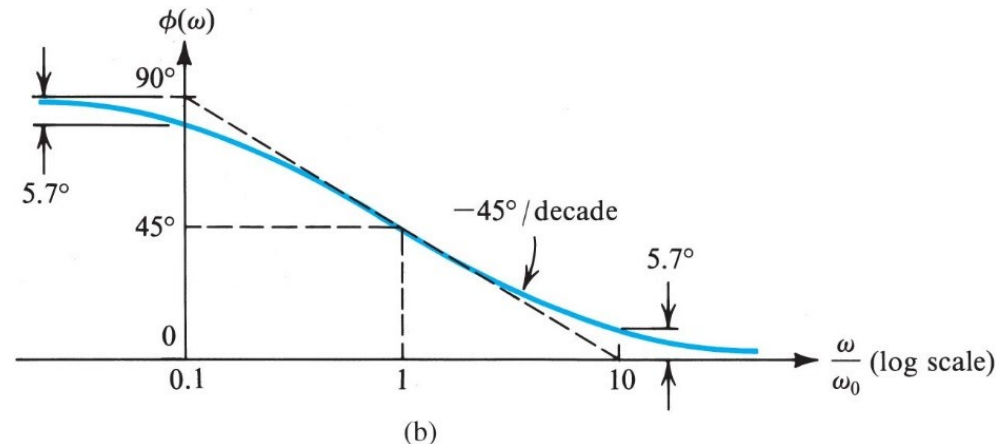


Figure 1.24 (a) Magnitude and (b) phase response of STC networks of the high-pass type.

Example: Amplifier Frequency Response

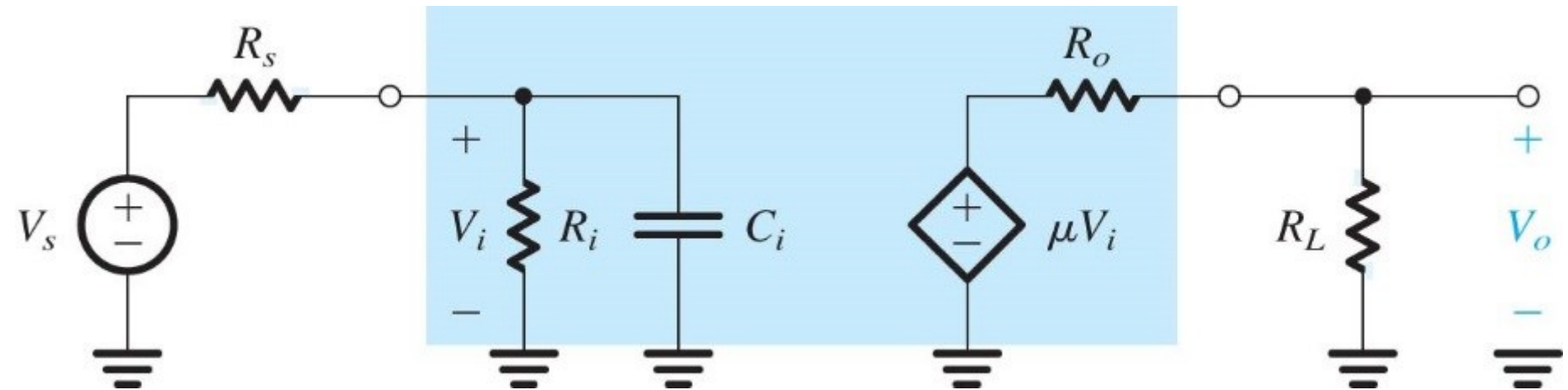
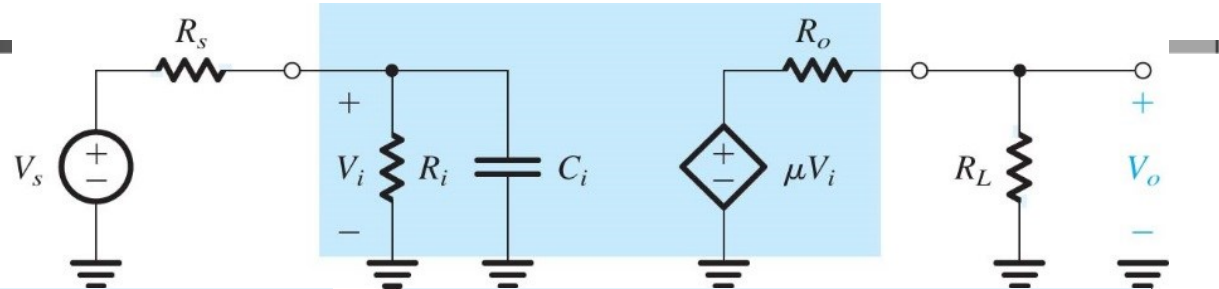


Figure 1.25 Circuit for Example 1.5.

Example: Amplifier Frequency Response



- (a) Derive an expression for the amplifier voltage gain V_o/V_s as a function of frequency. From this find expressions for the dc gain and the 3-dB frequency.
- (b) Calculate the values of the dc gain, the 3-dB frequency, and the frequency at which the gain becomes 0 dB (i.e., unity) for the case $R_s = 20 \text{ k}\Omega$, $R_i = 100 \text{ k}\Omega$, $C_i = 60 \text{ pF}$, $\mu = 144 \text{ V/V}$, $R_o = 200 \Omega$, and $R_L = 1 \text{ k}\Omega$.
- (c) Find $v_o(t)$ for each of the following inputs:
- $v_i = 0.1 \sin 10^7 t$, V
 - $v_i = 0.1 \sin 10^5 t$, V
 - $v_i = 0.1 \sin 10^6 t$, V
 - $v_i = 0.1 \sin 10^3 t$, V

Solution

- (a) Utilizing the voltage-divider rule, we can express V_i in terms of V_s as follows

$$V_i = V_s \frac{Z_i}{Z_i + R_s}$$

where Z_i is the amplifier input impedance. Since Z_i is composed of two parallel elements, it is obviously easier to work in terms of $Y_i = 1/Z_i$. Toward that end we divide the numerator and denominator by Z_i , thus obtaining

$$\begin{aligned} V_i &= V_s \frac{1}{1 + R_s Y_i} \\ &= V_s \frac{1}{1 + R_s [(1/R_i) + sC_i]} \end{aligned}$$

Thus,

$$\frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i) + sC_i R_s}$$

This expression can be put in the standard form for a low-pass STC network (see the top line of Table 1.2) by extracting $[1 + (R_s/R_i)]$ from the denominator; thus we have

$$\frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i)} \frac{1}{1 + sC_i [(R_s R_i)/(R_s + R_i)]} \quad (1.20)$$

At the output side of the amplifier we can use the voltage-divider rule to write

$$V_o = \mu V_i \frac{R_L}{R_L + R_o}$$

This equation can be combined with Eq. (1.20) to obtain the amplifier transfer function as

$$\frac{V_o}{V_s} = \mu \frac{1}{1 + (R_s/R_i)} \frac{1}{1 + (R_o/R_L)} \frac{1}{1 + sC_i [(R_s R_i)/(R_s + R_i)]} \quad (1.21)$$

Example 1.5 continued

We note that only the last factor in this expression is new (compared with the expression derived in the last section). This factor is a result of the input capacitance C_i , with the time constant being

$$\begin{aligned} \tau &= C_i \frac{R_s R_i}{R_s + R_i} \\ &= C_i (R_s \parallel R_i) \end{aligned} \quad (1.22)$$

We could have obtained this result by inspection: From Fig. 1.25 we see that the input circuit is an STC network and that its time constant can be found by reducing V_s to zero, with the result that the resistance seen by C_i is R_i in parallel with R_s . The transfer function in Eq. (1.21) is of the form $K/(1 + (s/\omega_0))$, which corresponds to a low-pass STC network. The dc gain is found as

$$K \equiv \frac{V_o}{V_s}(s=0) = \mu \frac{1}{1 + (R_s/R_i)} \frac{1}{1 + (R_o/R_L)} \quad (1.23)$$

The 3-dB frequency ω_0 can be found from

$$\omega_0 = \frac{1}{\tau} = \frac{1}{C_i (R_s \parallel R_i)} \quad (1.24)$$

Since the frequency response of this amplifier is of the low-pass STC type, the Bode plots for the gain magnitude and phase will take the form shown in Fig. 1.23, where K is given by Eq. (1.23) and ω_0 is given by Eq. (1.24).

- (b) Substituting the numerical values given into Eq. (1.23) results in

$$K = 144 \frac{1}{1 + (20/100)} \frac{1}{1 + (200/1000)} = 100 \text{ V/V}$$

Thus the amplifier has a dc gain of 40 dB. Substituting the numerical values into Eq. (1.24) gives the 3-dB frequency

$$\begin{aligned} \omega_0 &= \frac{1}{60 \text{ pF} \times (20 \text{ k}\Omega / 100 \text{ k}\Omega)} \\ &= \frac{1}{60 \times 10^{-12} \times (20 \times 100 / (20 + 100)) \times 10^3} = 10^6 \text{ rad/s} \end{aligned}$$

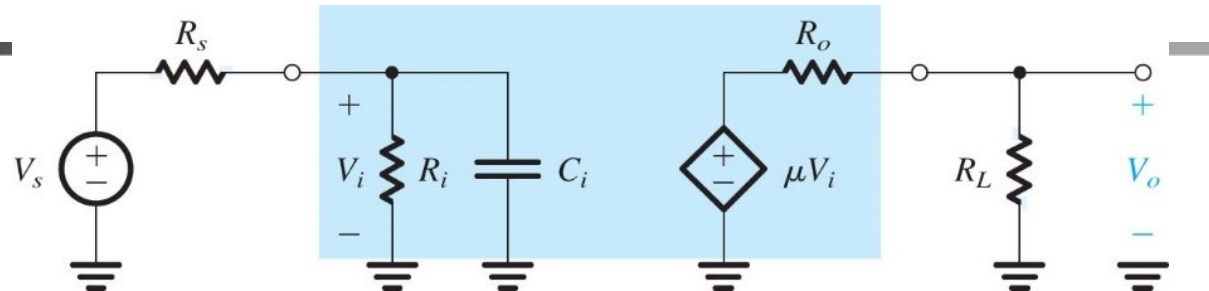
Thus,

$$f_0 = \frac{10^6}{2\pi} = 159.2 \text{ kHz}$$

Since the gain falls off at the rate of -20 dB/decade , starting at ω_0 (see Fig. 1.23a) the gain will reach 0 dB in two decades (a factor of 100); thus we have

$$\text{Unity-gain frequency} = 100 \times \omega_0 = 10^8 \text{ rad/s or } 15.92 \text{ MHz}$$

Example: Amplifier Frequency Response



- (c) To find $v_o(t)$ we need to determine the gain magnitude and phase at 10^2 , 10^5 , 10^6 , and 10^8 rad/s. This can be done either approximately utilizing the Bode plots of Fig. 1.23 or exactly utilizing the expression for the amplifier transfer function,

$$T(j\omega) \equiv \frac{V_o}{V_s}(j\omega) = \frac{100}{1 + j(\omega/10^6)}$$

We shall do both:

- (i) For $\omega = 10^2$ rad/s, which is $(\omega_0/10^4)$, the Bode plots of Fig. 1.23 suggest that $|T| = K = 100$ and $\phi = 0^\circ$. The transfer function expression gives $|T| \simeq 100$ and $\phi = -\tan^{-1} 10^{-4} \simeq 0^\circ$. Thus,

$$v_o(t) = 10 \sin 10^2 t, \text{ V}$$

- (ii) For $\omega = 10^5$ rad/s, which is $(\omega_0/10)$, the Bode plots of Fig. 1.23 suggest that $|T| \simeq K = 100$ and $\phi = -5.7^\circ$. The transfer function expression gives $|T| = 99.5$ and $\phi = -\tan^{-1} 0.1 = -5.7^\circ$. Thus,

$$v_o(t) = 9.95 \sin(10^5 t - 5.7^\circ), \text{ V}$$

- (iii) For $\omega = 10^6$ rad/s = ω_0 , $|T| = 100/\sqrt{2} = 70.7$ V/V or 37 dB and $\phi = -45^\circ$. Thus,

$$v_o(t) = 7.07 \sin(10^6 t - 45^\circ), \text{ V}$$

- (iv) For $\omega = 10^8$ rad/s, which is $(100 \omega_0)$, the Bode plots suggest that $|T| = 1$ and $\phi = -90^\circ$. The transfer function expression gives $|T| \simeq 1$ and $\phi = -\tan^{-1} 100 = -89.4^\circ$. Thus,

$$v_o(t) = 0.1 \sin(10^8 t - 89.4^\circ), \text{ V}$$

Typical Frequency Responses

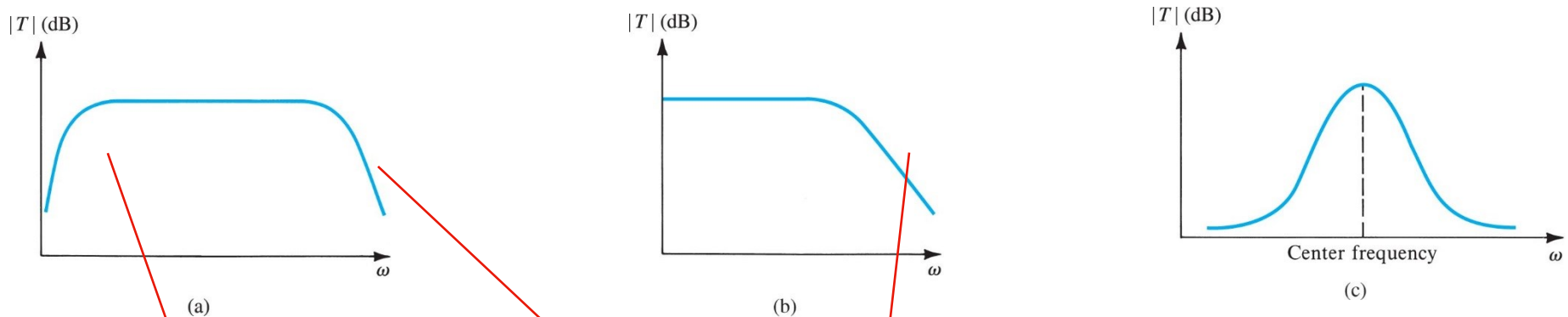


Figure 1.26 Frequency response for (a) a capacitively coupled amplifier, (b) a direct-coupled amplifier, and (c) a tuned or bandpass amplifier.

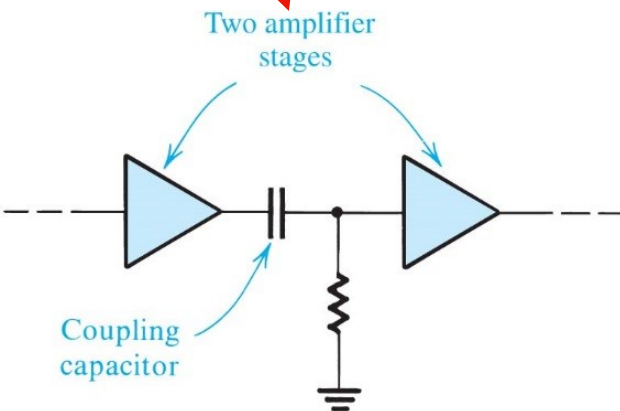


Figure 1.27 Use of a capacitor to couple amplifier stages.

High frequency cut-off due to **intrinsic capacitors** of the transistors

Low frequency roll-off due to **coupling capacitor**